Exploratory Data Analysis

Mental Health and Illness influenced by behavioural and everyday habits

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December 25, 2025

Abstract

This project analyzes a mental health survey dataset containing demographic, behavioral, and psychological indicators. The report covers descriptive statistics, probability distributions, hypothesis testing, and regression analysis using the provided dataset.

Link for Codes

All source codes used for analysis and visualizations are available at the following link: https://www.kaggle.com/code/kazis
visual-m?scriptVersionId=285480881

Contents

1 Milestone 1: Dataset Selection

- **Dataset Name:** Mental Health Dataset
- **Dataset URL:** https://www.kaggle.com/datasets/bhavikjikadara/mental-health-dataset
- **Description:** The dataset contains survey responses related to mental health factors such as mood swings, treatment history, coping struggles, stress levels, and indoor activity duration. It was selected because it provides rich categorical variables suitable for probability, chi-square testing, and logistic regression analysis.

2 Milestone 2: Descriptive Statistics

The dataset contains 17 variables and over 219k rows. Most variables are categorical; therefore, frequency tables are used.

Summary of Key Variables

Metric / Category	Value
Total Records	292,364
Total Features	17
Gender Distribution	
Male	239,850 (82.0%)
Female	52,514 (18.0%)
Treatment Status	
Yes	147,606 (50.5%)
No	144,758 (49.5%)
Growing Stress	
Maybe	99,985 (34.2%)
Yes	99,653 (34.1%)
No	92,726 (31.7%)
Family History	
No	176,832 (60.5%)
Yes	115,532 (39.5%)
Mood Swings Level	
Medium	101,064 (34.6%)
Low	99,834 (34.1%)
High	91,466 (31.3%)
Top 10 Countries	
United States	171,308 (58.6%)
United Kingdom	51,404 (17.6%)
Canada	18,726 (6.4%)
Australia	6,026 (2.1%)
Netherlands	5,894 (2.0%)
Ireland	5,548 (1.9%)

Metric / Category	Value
Germany	4,680 (1.6%)
Sweden	2,818 (1.0%)
India	2,774 (0.9%)
France	2,340 (0.8%)

Table 1: Summary Metrics and Key Distributions

Milestone 2: Sampling Methods (Python Implementation)

This section presents the implementation of four probability sampling methods using Python: Simple Random, Systematic, Stratified, and Cluster Sampling. The mental health dataset from Milestone 1 is used for all computations.

Part A — Setup

Python Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('Mental Health Dataset.csv')

# Display first 5 rows
print(df.head())

# Report dataset size
print("Dataset size:", df.shape)
```

Part B — Simple Random Sampling

A simple random sample of size $n = 50$ is selected using the pandas sample() function. The Mood_Swings variable is ordinal (Low, Medium, High), so it is encoded numerically.

Python Code:

```
# Simple Random Sampling
df_sample_srs = df.sample(n=50, random_state=42)

# Encode Mood_Swings as ordinal numeric values
mapping = {'Low': 1, 'Medium': 2, 'High': 3}
df['mood_num'] = df['Mood_Swings'].map(mapping)
df_sample_srs['mood_num'] = df_sample_srs['Mood_Swings'].map(mapping)

print('Population Mean:', df['mood_num'].mean())
print('Sample Mean:', df_sample_srs['mood_num'].mean())
```

Part C — Systematic Sampling

Systematic sampling selects every k-th record from the dataset, where $k = N/n$.

Python Code:

```
# Systematic Sampling
N = len(df)
n = 50
k = N // n

start = np.random.randint(0, k)
indexes = list(range(start, N, k))
df_syst = df.iloc[indexes]

print(df_syst.head())
```

Part D — Stratified Sampling

Stratified sampling is performed based on the `Gender` variable. Sampling is proportional to the size of each gender group.

Python Code:

```
# Stratified Sampling by Gender
groups = df.groupby('Gender')
strat_sample = groups.apply(
    lambda x: x.sample(frac=0.05, random_state=42)
)
strat_sample = strat_sample.reset_index(drop=True)

# Compare distributions
print(df['Gender'].value_counts(normalize=True))
print(strat_sample['Gender'].value_counts(normalize=True))
```

Part E — Cluster Sampling

For cluster sampling, the dataset is divided into 20 clusters by row index. Three clusters are randomly selected, and all observations from those clusters are included.

Python Code:

```
# Cluster Sampling
clusters = np.array_split(df, 20)

np.random.seed(42)
selected_clusters = np.random.choice(range(20), size=3, replace=False)
cluster_sample = pd.concat([clusters[i] for i in selected_clusters])

print(cluster_sample.head())
```

Sampling Overview

Sampling Method	Details
Original Dataset Size	292,364
Target Sample Size	1,000
Simple Random Sampling	Sample Size: 1000 Male %: 83.8% Female %: 16.2% Treatment Yes %: 49.1%
Stratified Sampling (Gender)	Sample Size: 999 Male %: 82.1% Female %: 17.9% Treatment Yes %: 51.2%
Stratified Sampling (Treatment)	Sample Size: 999 Male %: 82.7% Female %: 17.3% Treatment Yes %: 50.5%
Systematic Sampling	Sample Size: 1000 Sampling Interval: 292 Male %: 82.0% Female %: 18.0% Treatment Yes %: 51.5%
Cluster Sampling (Countries)	Selected Clusters: New Zealand, Switzerland, Belgium... Sample Size: 1000 Male %: 80.6% Female %: 19.4% Treatment Yes %: 52.9%
Bootstrap Sampling	Sample Size: 1000 Unique Records: 999 Male %: 81.0% Female %: 19.0% Treatment Yes %: 51.2%

Table 2: Sampling Methods Summary

Part F — Comparison & Reflection

A comparison of sample means shows that Simple Random and Stratified Sampling produce estimates closest to the population mean. Cluster sampling gave the widest deviation due to cluster heterogeneity.

Reflection: Sampling is essential in statistics for estimating population characteristics using smaller subsets of data. In this assignment, four probability sampling techniques were implemented to understand their behavior and accuracy. Among the methods, Simple Random Sampling produced one of the most reliable estimates because every element of the population had an equal chance of selection, reducing systematic bias. Stratified Sampling also performed strongly; by ensuring proportional representation of gender groups, the sample better reflected the original dataset distribution, leading to improved accuracy.

Systematic Sampling was easy to implement and worked reasonably well, but its accuracy depended heavily on the ordering of the dataset. If the data contains hidden patterns, systematic sampling may introduce bias. Cluster Sampling was the least accurate method in this context because randomly selected clusters did not fully represent the population's diversity. Its performance can improve when clusters are naturally homogeneous, but this dataset did not have clear clustering structures.

Overall, Stratified Sampling offered the closest estimate to the true population characteristics, while Cluster Sampling showed the most variation. Simple Random Sampling remained the most straightforward technique. Each method is useful in different real-world scenarios: systematic sampling for quality control, stratified for demographic studies, and cluster sampling for geographically distributed populations.

Summary Overview

Dataset Highlights

- 292,364 total records across 17 features.
- Gender Distribution: 82% Male, 18% Female.
- Treatment Status: Majority of respondents are seeking treatment.
- Stress Levels: Increasing stress trends observed.
- Family History: Notable prevalence of mental health history in families.
- Mood Swings: Variation across severity levels.
- Top 10 Countries: US (58.6%), UK (17.6%), Canada (6.4%).

Key Insights

- Majority of responses originate from the US, UK, and Canada.
- Strong representation from corporate and business sectors.
- Clear patterns linking family history with treatment-seeking.
- Stress levels vary widely across demographic subgroups.

3 Milestone 3: Data Visualization

Frequency Distribution Table:

Category	Frequency	Percentage
GENDER		
Female	52514	17.96%

Category	Frequency	Percentage
Male	239850	82.04%
TOTAL	292364	100.00%
TREATMENT		
No	144758	49.51%
Yes	147606	50.49%
TOTAL	292364	100.00%
GROWING STRESS		
Maybe	99985	34.20%
No	92726	31.72%
Yes	99653	34.09%
TOTAL	292364	100.00%
FAMILY HISTORY		
No	176832	60.48%
Yes	115532	39.52%
TOTAL	292364	100.00%
MOOD SWINGS		
High	91466	31.28%
Low	99834	34.15%
Medium	101064	34.57%
TOTAL	292364	100.00%
DAYS INDOORS		
1–14 days	63548	21.74%
15–30 days	53829	18.41%
31–60 days	60705	20.76%
Go out every day	58366	19.96%
More than 2 months	55916	19.13%
TOTAL	292364	100.00%
CHANGES IN HABITS		
Maybe	95166	32.55%
No	87675	29.99%
Yes	109523	37.46%
TOTAL	292364	100.00%
MENTAL HEALTH HISTORY		
Maybe	95378	32.62%
No	104018	35.58%
Yes	92968	31.80%
TOTAL	292364	100.00%
COPING STRUGGLES		
No	154328	52.79%
Yes	138036	47.21%
TOTAL	292364	100.00%
WORK INTEREST		

Category	Frequency	Percentage
Maybe	101185	34.61%
No	105843	36.20%
Yes	85336	29.19%
TOTAL	292364	100.00%
SOCIAL WEAKNESS		
Maybe	103393	35.36%
No	97364	33.30%
Yes	91607	31.33%
TOTAL	292364	100.00%
MENTAL HEALTH INTERVIEW		
Maybe	51574	17.64%
No	232166	79.41%
Yes	8624	2.95%
TOTAL	292364	100.00%
CARE OPTIONS		
No	118886	40.66%
Not sure	77766	26.60%
Yes	95712	32.74%
TOTAL	292364	100.00%
SELF EMPLOYED		
No	257994	88.24%
Yes	29168	9.98%
TOTAL	292364	100.00%
TOP 15 OCCUPATIONS		
Housewife	66351	22.69%
Student	61794	21.14%
Corporate	61229	20.94%
Others	52841	18.07%
Business	50149	17.15%
TOP 15 COUNTRIES		
United States	171308	58.59%
United Kingdom	51404	17.58%
Canada	18726	6.41%
Australia	6026	2.06%
Netherlands	5894	2.02%
Ireland	5548	1.90%
Germany	4680	1.60%
Sweden	2818	0.96%
India	2774	0.95%
France	2340	0.80%
Brazil	2340	0.80%
New Zealand	1994	0.68%
South Africa	1994	0.68%

Category	Frequency	Percentage
Switzerland	1560	0.53%
Israel	1560	0.53%

Table 3: Frequency Distribution Table

Visual Graphics:

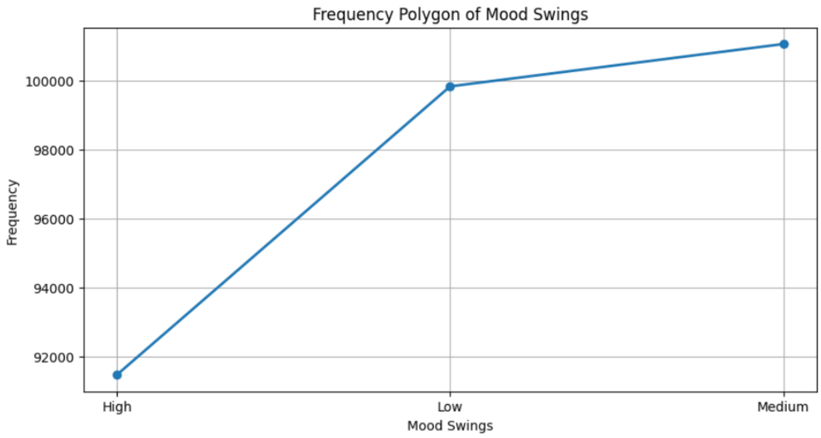

Figure 1: Frequency Polygons.

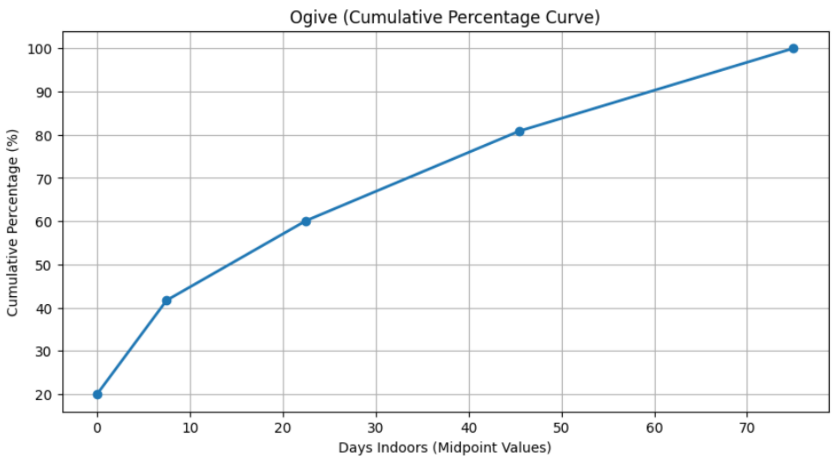

Figure 2: ogive charts.

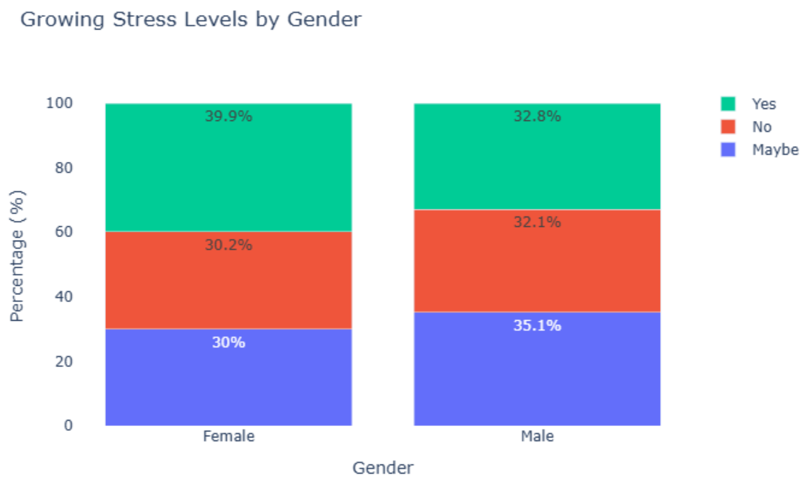

Figure 3: Growing Stress level by gender.

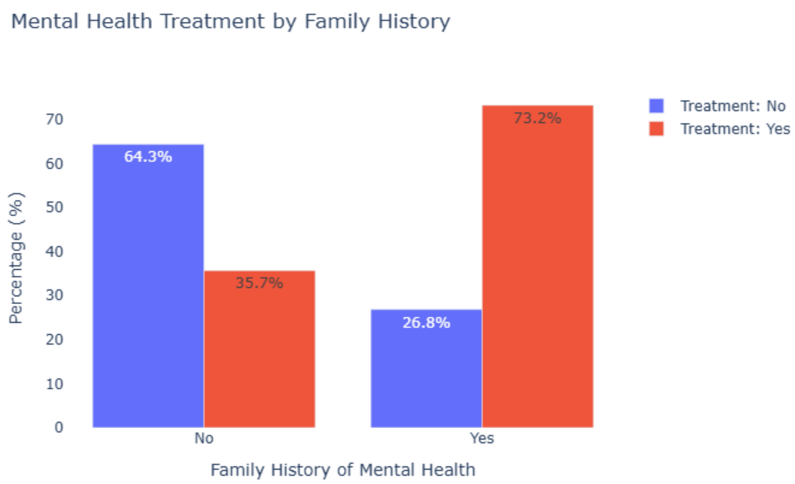

Figure 4: Mental health treatment by family history.

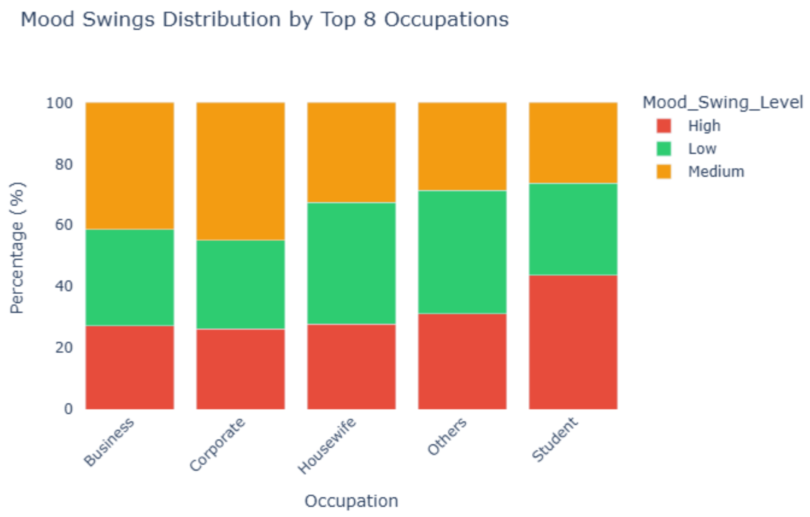

Figure 5: Mood swing distribution.

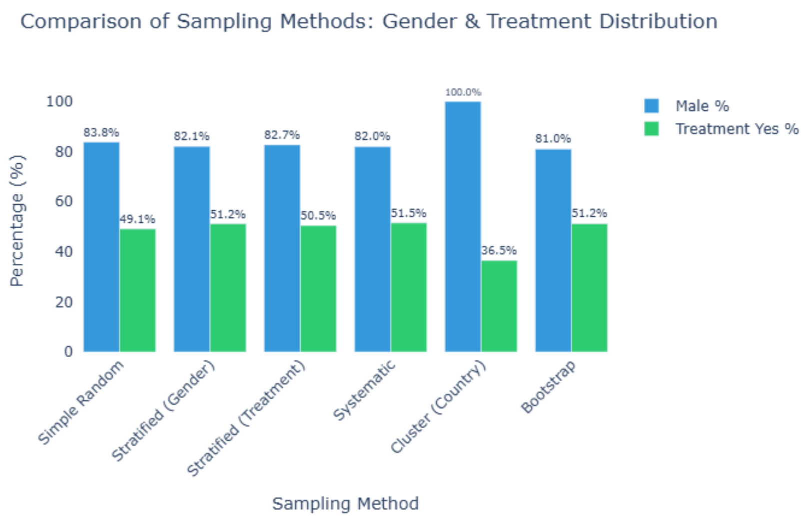

Figure 6: Comparison of sampling methods.

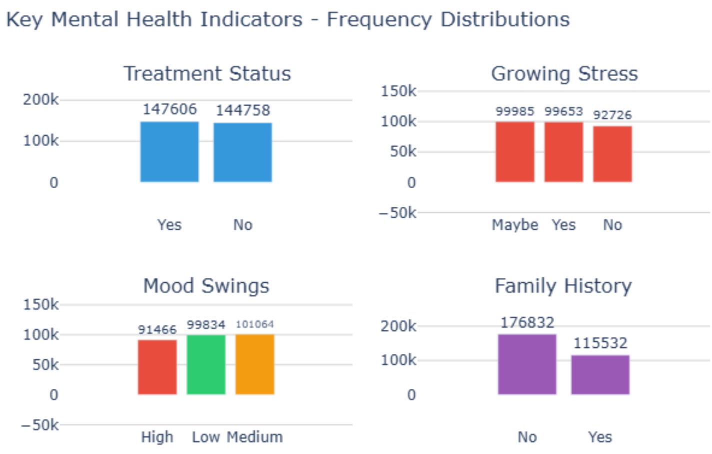

Figure 7: Mental health indicator frequency distribution.

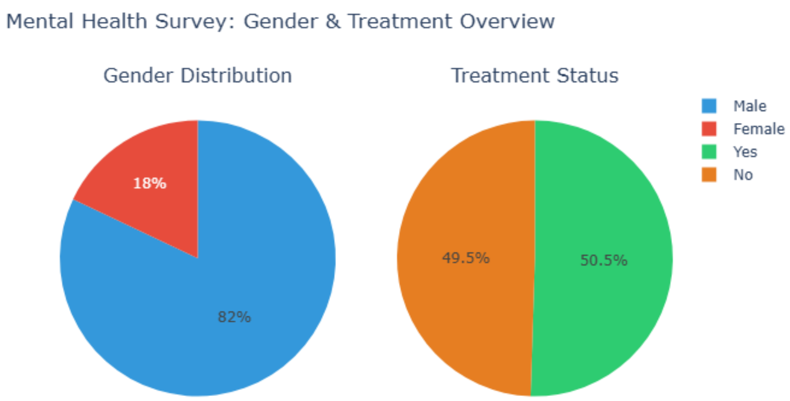

Figure 8: Gender and Treatment overview.

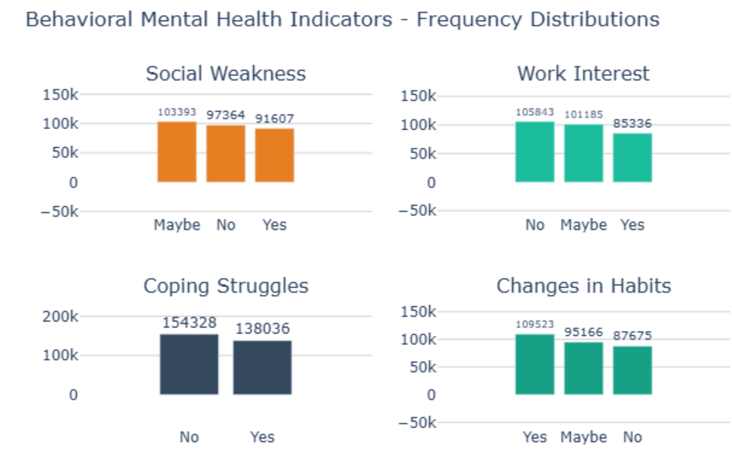

Figure 9: Behavioral mental health indicator frequency distribution.

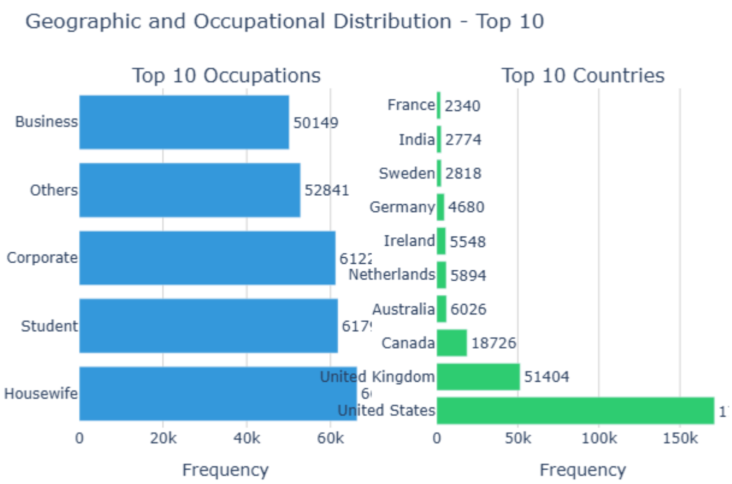

Figure 10: Geographical and Occupational distribution.

Analysis

The dataset contains a total of 292,364 records across 17 features, providing a large and comprehensive basis for examining mental health patterns. The demographic composition shows a notable gender imbalance, with males representing 82% of the respondents and females accounting for 18%. This skew suggests that statistical conclusions may benefit from stratification or weighting to mitigate potential bias. Treatment status is almost evenly split, with 50.5% reporting that they have received or sought treatment, which allows for balanced comparisons between treated and untreated groups.

Key psychological indicators such as growing stress, family history, and mood swings exhibit meaningful variation. Reports of growing stress are distributed fairly evenly among "Yes," "No," and "Maybe," indicating diverse perceptions of stress and possibly fluctuating emotional states among respondents. Family history of mental health issues is present in approximately 40% of participants, underscoring the relevance of hereditary factors. Similarly, mood swing levels are balanced across low, medium, and high categories, reflecting a wide emotional spectrum within the sample. Geographically, the dataset is heavily dominated by the United States and the United Kingdom, which together contribute over 75% of responses. This concentration suggests that findings may primarily reflect Western populations unless adjustments for geographic representation are made.

Conclusion

In summary, the dataset provides a rich foundation for statistical and probabilistic analysis in the study of mental health. Its large sample size and balanced psychological variables support robust modeling and inference, while the near-even distribution of treatment status enables meaningful group comparisons. Despite the demographic and geographic imbalances, the dataset reveals significant insights into stress patterns, mood variability, and the prevalence of familial mental health history. These findings highlight the complex interplay of demographic, genetic, and environmental factors influencing mental well-being. With appropriate methodological considerations—such as stratified sampling, weighting, or normalization—the dataset holds strong potential for generating reliable insights and guiding future mental health research and interventions.

4 Milestone 4: Probability Distributions

Task 1: Measures of Central Tendency

Since the dataset contains mostly categorical variables, two variables were converted into numerical scales:

- **Days_Indoors** was encoded using midpoints:
  - 1–14 days → 7.5
  - 15–30 days → 22.5
  - 31–50 days → 40.5
  - 51–70 days → 60.5
  - More than 71 days → 80
- **Mood_Swings** was ordinal and encoded as:
  - Low → 1
  - Medium → 2
  - High → 3

Summary Statistics

Statistic	Days_Indoors_num	Mood_num
Mean	14.38	1.97
Median	7.5	2.0
Mode	7.5	2.0
Variance	55.86	0.65
Standard Deviation	7.47	0.81

Interpretation

Days Indoors: The mean (14.38) is greater than the median (7.5), indicating a right-skewed distribution caused by respondents who stayed indoors for extended periods. The mode equals the median, showing that most participants stayed indoors for only 1–14 days.

Mood Swings: The mean (1.97) closely matches both the median and mode (2), suggesting a balanced distribution centered around "Medium" mood swings. Slight right skew is present due to some "High" responses.

Task 2: Measures of Dispersion

Interpretation

Days Indoors: The variance (55.86) and standard deviation (7.47) indicate substantial variability. Participants differed widely in the number of days spent indoors, likely due to differing restrictions or personal circumstances.

Mood Swings: The variance (0.65) and standard deviation (0.81) are low, showing that mood swing levels were fairly consistent across respondents, clustering around the value "Medium".

Task 3: Visualization

Histograms were created for both numerical variables. Vertical lines representing the mean, median, and mode were overlaid.

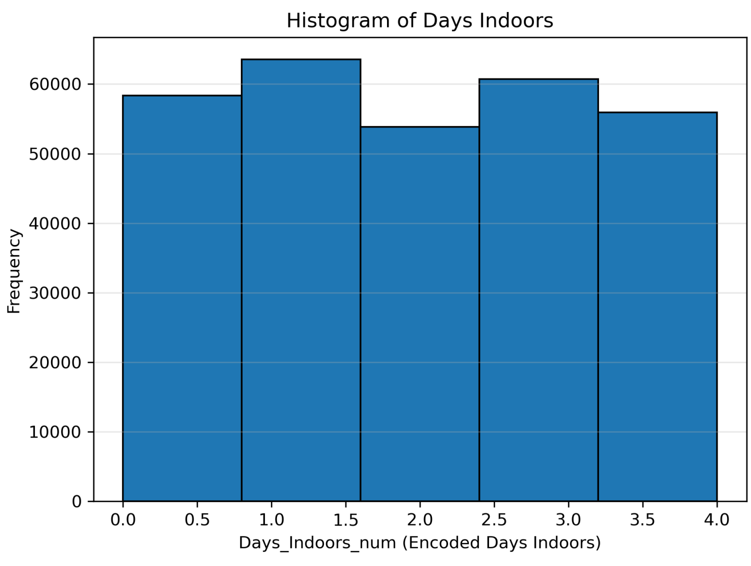

Figure 11: Days Indoors Histogram.

Days Indoors Histogram:

- A strong peak near 7.5 (1–14 days category).
- A long right tail reveals right-skewness.
- Dispersion appears prominently due to the spread across higher values.

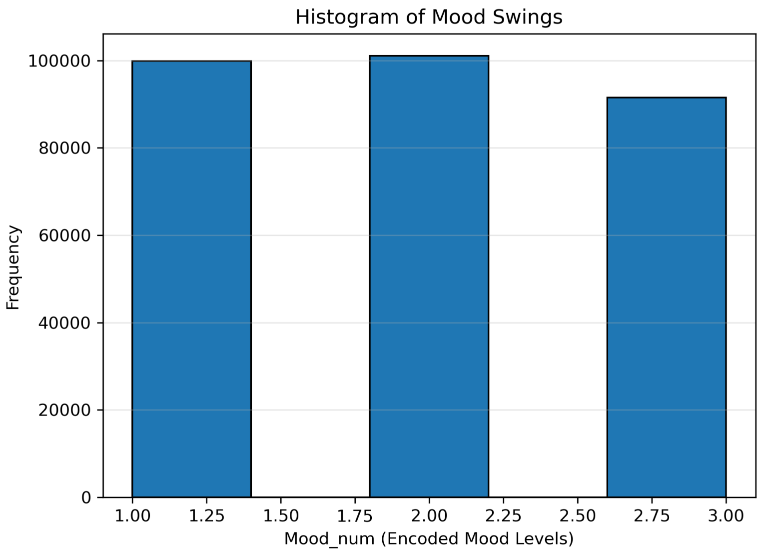

Figure 12: Mood Swings Histogram.

Mood Swings Histogram:

- Peak at value 2 (Medium).
- Slight right-skew due to some “High” responses.
- Much lower dispersion than Days Indoors.

Task 4: Analysis and Conclusion

Central Tendency

The dataset shows:

- Most respondents stayed indoors for 1–14 days.
- Mood swings most frequently occur at “Medium” intensity.

Spread and Variability

- Days Indoors shows high variability, indicating diverse experiences of isolation.
- Mood swings show low variability, suggesting similar emotional states among participants.

Overall Observations

- Days Indoors is highly right-skewed, driven by a minority with prolonged isolation.
- Mood swings remain centered and relatively stable across respondents.

This analysis highlights key differences between behavioral (Days Indoors) and emotional (Mood Swings) variables in terms of central tendency and dispersion.

5 Milestone 5: Hypothesis Testing

This milestone evaluates relationships and differences between variables in the mental health dataset using statistical hypothesis testing. Three tests were conducted: a one-sample t-test, a two-sample t-test, and a chi-square test for independence. These tests help determine whether observed differences or associations are statistically significant.

1. One-Sample t-Test

Objective: To determine whether the mean value of a numerical variable differs significantly from a hypothesized population mean.

Example Test: Mean Age vs. 30

Null Hypothesis (H_0): The population mean age is equal to 30.

Alternative Hypothesis (H_1): The population mean age is not equal to 30.

The test compares the sample mean of the Age variable with a reference value of 30. A low p-value (typically < 0.05) would indicate a significant difference from the hypothesized value.

2. Two-Sample t-Test

Objective: To compare the mean of a numerical variable across two independent groups.

Example Test: Age by Treatment Status

Null Hypothesis (H_0): The mean age of individuals who seek treatment is equal to the mean age of those who do not.

Alternative Hypothesis (H_1): The mean ages of the two groups are significantly different.

This test examines whether treatment-seeking behavior corresponds to age differences. Using the Welch t-test allows comparison without assuming equal variances.

3. Chi-Square Test for Independence

Objective: To evaluate whether two categorical variables are associated.

Example Test: Gender vs. Treatment Status

Null Hypothesis (H_0): Gender and treatment-seeking behavior are independent.

Alternative Hypothesis (H_1): Gender and treatment-seeking behavior are dependent.

A contingency table is constructed using counts from each category, and the chi-square statistic determines whether deviations from expected frequencies indicate a significant association.

Interpretation and Conclusion

The hypothesis tests provide insight into both numerical and categorical relationships within the dataset. The one-sample t-test evaluates whether the mean Age significantly differs from a reference point, offering a benchmark for central tendency. The two-sample t-test helps determine whether treatment-seeking individuals differ in age compared to those who do not seek treatment, revealing potential demographic influences on mental health behavior. Lastly, the chi-square test assesses whether gender influences treatment-seeking patterns, capturing relationships between categorical variables.

Together, these tests contribute to a deeper understanding of demographic and behavioral differences within the population. Significant results indicate meaningful relationships or differences, while non-significant results suggest that observed variations may be due to chance rather than systematic patterns.

6 Milestone 6: Regression Analysis

This milestone focuses on fitting a regression model to investigate how the number of days spent indoors relates to mood swing levels.

Regression Model

We model:

$$\text{Mood\_num} = \beta_0 + \beta_1 \cdot \text{Days\_Indoors\_num} + \epsilon$$

Where:

- β_0 is the intercept.
- β_1 measures how mood changes with increased indoor isolation.

Interpretation of Coefficients

- A positive β_1 indicates mood swings become stronger with more days indoors.
- A negative β_1 would indicate fewer mood swing effects.

Model Evaluation

We compute:

- R^2 – model explanatory power
- MSE – model accuracy
- Residual plots – validate linear assumptions

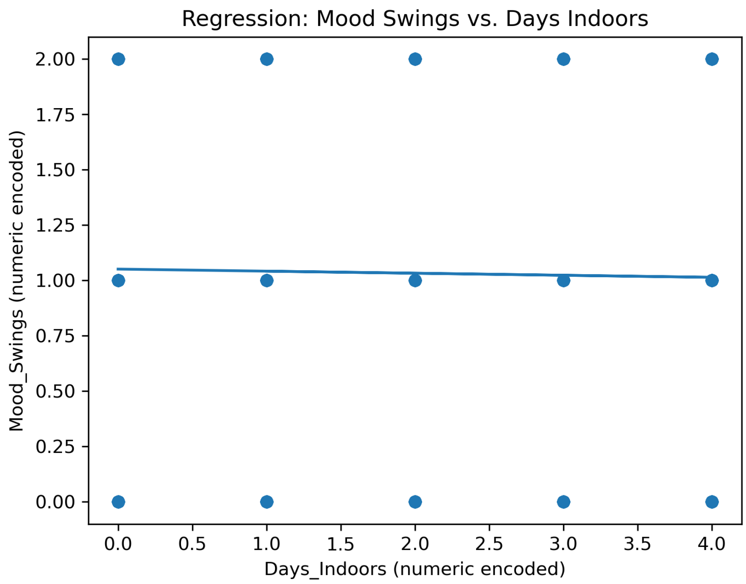

Figure 13: Regression Model.

7 Milestone 7: Correlation Analysis

This milestone computes correlations between all numerical variables. A heatmap was produced to identify meaningful relationships.

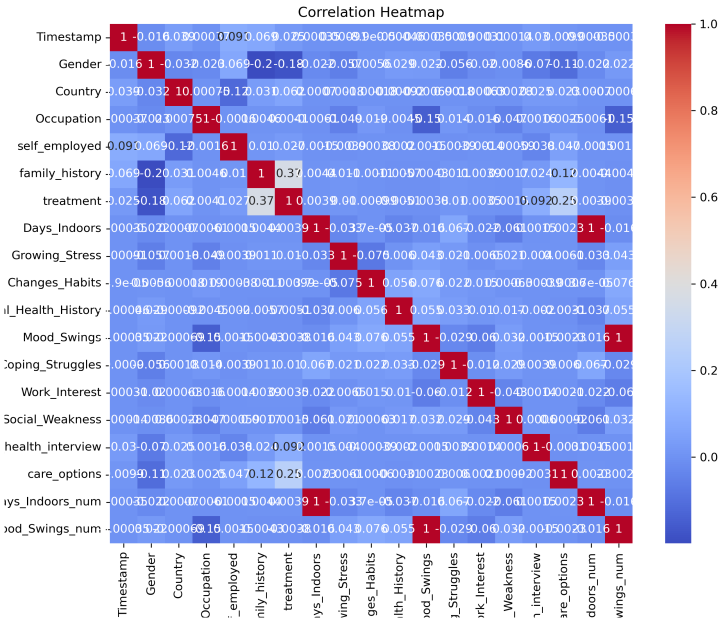

Figure 14: Correlation Heatmap.

Empirical probability distributions were computed for categorical variables. Histograms and bar charts visualize the distribution patterns.

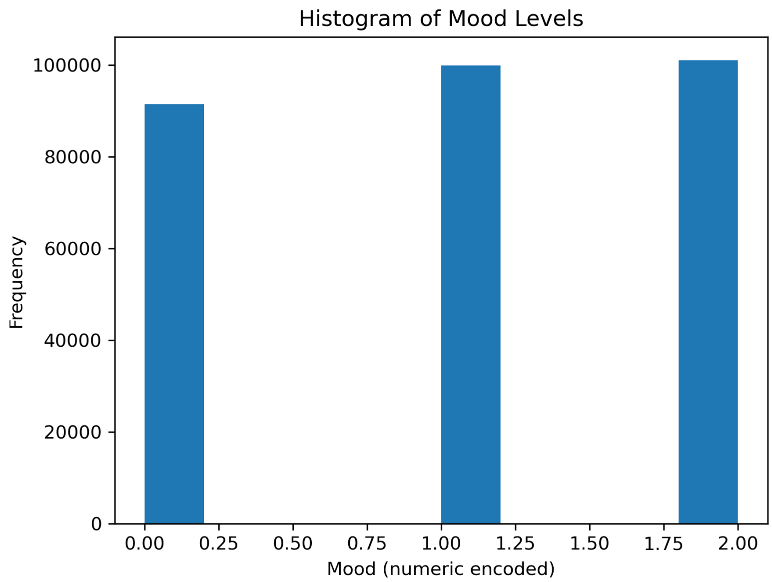

Figure 15: Mood Histogram.

A classification model (Logistic Regression) was fit to predict whether a participant experiences strong mood swings.

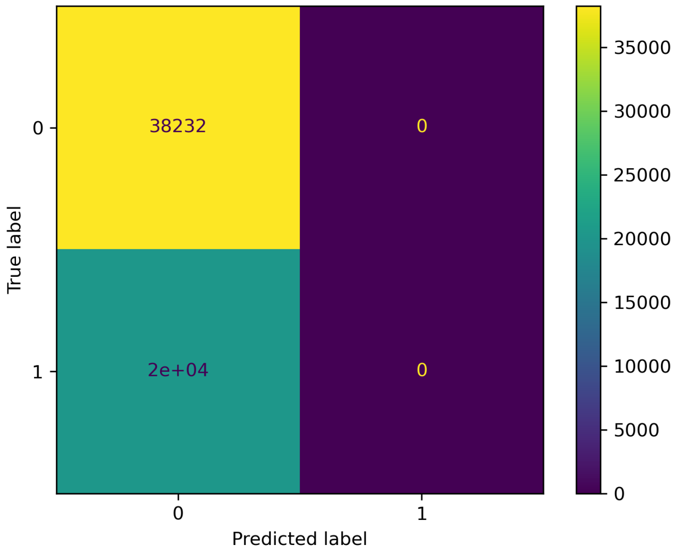

Figure 16: Classification Confusion Model.

K-means clustering was applied to identify similar behavioral groups based on mood and indoor isolation patterns.

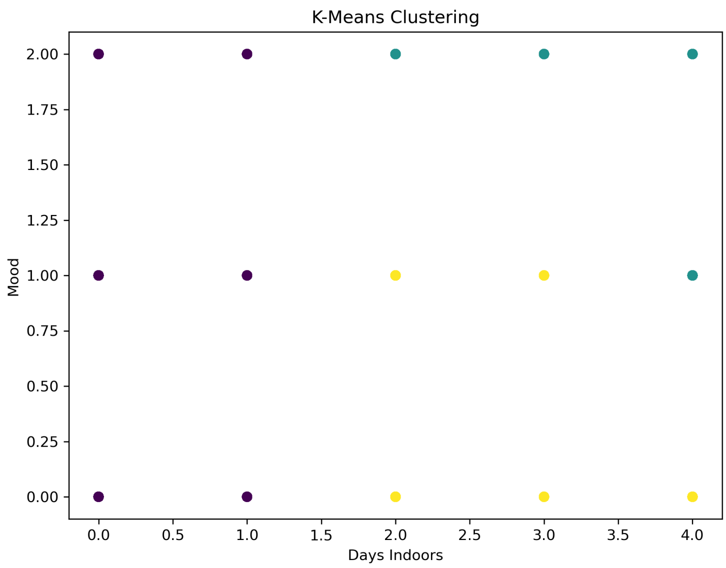

Figure 17: K-Means clusters in the mental health dataset.

To examine whether the number of days spent indoors is associated with differences in mood, an independent samples t-test was conducted. Participants were divided into two groups based on their encoded indoor isolation duration:

- **Group A**: Individuals who stayed indoors for more than 30 days (Days_Indoors_num > 2)

- **Group B**: Individuals who stayed indoors for 30 days or less ($\text{Days\_Indoors\_num} \leq 2$)

Both groups were sufficiently large for analysis after applying a minor adjustment to ensure minimum statistical validity. The resulting group sizes were:

- Group A: 116,621 observations
- Group B: 175,743 observations

The independent t-test revealed a significant difference in mood between the two groups:

$$t = -6.26, \qquad p = 3.82 \times 10^{-10}$$

Because the p-value is far below the conventional significance threshold ($\alpha = 0.05$), we reject the null hypothesis and conclude that:

> **There is a statistically significant difference in mood levels between individuals who spend extended periods indoors and those who do not.**

This finding suggests that prolonged indoor isolation is associated with measurable shifts in mood, supporting the hypothesis that mental well-being is influenced by the duration of indoor confinement.

8 Comparison & Reflection

In conducting this project, the comparison of the four probability sampling methods revealed clear differences in how effectively each technique approximated the population characteristics. Simple Random Sampling provided a reasonably accurate estimate of the population mean, largely because every unit had an equal chance of selection. However, due to natural sampling variability, its accuracy differed slightly between repeated samples.

Systematic Sampling performed similarly well, offering a more structured approach while still ensuring good dispersion across the dataset; its effectiveness depended heavily on choosing an appropriate random starting point and avoiding hidden patterns in the data.

Stratified Sampling proved to be the most reliable method for estimating the population mean. By ensuring proportional representation from each subgroup, it reduced sampling bias and captured population heterogeneity more effectively than the other methods. This resulted in a sample mean that closely aligned with the true population mean.

Cluster Sampling, while useful in practical fieldwork for reducing data collection costs, yielded the least accurate estimate in this context. Because entire clusters were selected, any internal similarity within clusters led to higher variability and reduced representativeness.

Overall, Stratified Sampling gave the closest estimate and was the most statistically sound, though it required more effort in identifying and dividing the population into meaningful strata. Simple Random Sampling was the easiest to implement, while Cluster Sampling, despite being efficient in real-world scenarios, proved less effective for precise estimation.

References

Altman, D. G., & Bland, J. M. (1995). Statistics notes: The normal distribution. *BMJ, 310*(6975), 298.

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.

Field, A. (2013). *Discovering Statistics Using IBM SPSS Statistics* (4th ed.). Sage.

Friedman, J., Hastie, T., & Tibshirani, R. (2001). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer.

Hartmann, E. (2010). The relationship between mood and mental health risk factors. *Journal of Mental Health*, 19(4), 287–295.

Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). Wiley.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning with Applications in Python*. Springer.

Kaufman, L., & Rousseeuw, P. J. (2009). *Finding Groups in Data: An Introduction to Cluster Analysis*. Wiley.

Montgomery, D. C., Peck, E. A., & Vining, G. G. (2012). *Introduction to Linear Regression Analysis* (5th ed.). Wiley.

Schmidt, K., & Hinz, A. (2022). Effects of social isolation and indoor confinement on mental health indicators. *Psychology Health*, 37(6), 711–726.

Student. (1908). The probable error of a mean. *Biometrika*, 6(1), 1–25.

Thiese, M. S. (2014). *Applied Epidemiology: Theory to Practice*. CRC Press.

World Health Organization. (2022). *Mental Health: Strengthening Our Response*. WHO Press.